```
****************************************************
*                                                  *
*           STAAD.Pro                              *
*           Version  2007    Build 04              *
*           Proprietary Program of                 *
*           Research Engineers,  Intl.             *
*           Date=    JAN 20, 2013                  *
*           Time=    10:10:55                      *
*                                                  *
*      USER ID:                                    *
****************************************************
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```
     1. STAAD SPACE
INPUT FILE: 20121120 T-Goal Post 045 FR1.STD
     2. START JOB INFORMATION
     3. ENGINEER DATE 12-MAY-12
     4. END JOB INFORMATION
     5. INPUT WIDTH 79
     6. UNIT METER KN
     7. JOINT COORDINATES
     8. 2 5 1 -5.73; 4 5 4.495 -5.73; 5 5 4.495 -4.98; 6 5 4.495 -6.48
     9. MEMBER INCIDENCES
    10. 3 2 4; 4 4 5; 5 6 4
    11. DEFINE MATERIAL START
    12. ISOTROPIC STEEL
    13. E 2.05E+008
    14. POISSON 0.3
    15. DENSITY 76.8195
    16. ALPHA 1.2E-005
    17. DAMP 0.03
    18. END DEFINE MATERIAL
    19. MEMBER PROPERTY JAPANESE
    20. 4 5 TABLE ST H428X407X20
    21. 3 TABLE ST H428X407X20
    22. CONSTANTS
    23. BETA 90 MEMB 3
    24. MATERIAL STEEL ALL
    25. SUPPORTS
    26. 2 FIXED
    27. **************************** DEAD LOAD *****************************
    28. LOAD 1 LOADTYPE NONE  TITLE DL
    29. SELFWEIGHT Y -1
    30. MEMBER LOAD
    31. 5 CON GY -19.6 0.35
    32. **************************** FRICTIONAL LOAD *****************************
    33. LOAD 2 LOADTYPE NONE  TITLE FL
    34. MEMBER LOAD
    35. 5 CON GX -5.88 0.35
    36. **************************** WIND LOAD X-DIRECTION*****************************
    37. LOAD 3 LOADTYPE NONE  TITLE WLX
    38. MEMBER LOAD
    39. 4 5 UNI GX -0.364
    40. 3 UNI GX -0.364
    41. **************************** WIND LOAD Z-DIRECTION*****************************
    42. LOAD 4 LOADTYPE NONE  TITLE WLZ
    43. MEMBER LOAD
    44. 3 UNI GZ -0.346
    45. **************************** NOTIONAL LOAD X-DIRECTION*************************
    46. LOAD 5 LOADTYPE NONE  TITLE NLX
    47. SELFWEIGHT X -0.015
    48. MEMBER LOAD
    49. 5 CON GX -0.294 0.35
    50. ****************************NOTIONAL LOAD Z-DIRECTION**************************
    51. LOAD 6 LOADTYPE NONE  TITLE NLZ
    52. SELFWEIGHT Z -0.015
    53. MEMBER LOAD
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    54. 5 CON GZ -0.294 0.35
    55. ************* LOAD COMBINATION - SERVICE *********
    56. LOAD COMB 7 1.0DL+1.0FL
    57. 1 1.0 2 1.0
    58. LOAD COMB 8 1.0DL+1.0FL+1.0WLX
    59. 1 1.0 2 1.0 3 1.0
    60. LOAD COMB 9 1.0DL+1.0FL+1.0WLX
    61. 1 1.0 2 1.0 4 1.0
    62. LOAD COMB 10 1.0DL+1.0FL+1.0NLX
    63. 1 1.0 2 1.0 5 1.0
    64. LOAD COMB 11 1.0DL+1.0FL+1.0NLZ
    65. 1 1.0 2 1.0 6 1.0
    66. ************ LOAD COMBINATION - ULTIMATE ********
    67. LOAD COMB 12 1.0DL+1.4FL
    68. 1 1.0 2 1.4
    69. LOAD COMB 13 1.4DL+1.4FL
    70. 1 1.4 2 1.4
    71. LOAD COMB 14 1.0DL+1.4FL+1.4WLX
    72. 1 1.0 2 1.4 3 1.4
    73. LOAD COMB 15 1.0DL+1.4FL+1.4WLZ
    74. 1 1.0 2 1.4 4 1.4
    75. LOAD COMB 16 1.0DL+1.4FL+1.4NLX
    76. 1 1.0 2 1.4 5 1.4
    77. LOAD COMB 17 1.0DL+1.4FL+1.4NLZ
    78. 1 1.0 2 1.4 6 1.4
    79. PERFORM ANALYSIS
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P R O B L E M   S T A T I S T I C S

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS =   4/   3/   1

SOLVER USED IS THE IN-CORE ADVANCED SOLVER

TOTAL PRIMARY LOAD CASES =   6, TOTAL DEGREES OF FREEDOM =   18

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    80. LOAD LIST 7 TO 11
    81. PRINT JOINT DISPLACEMENTS ALL
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JOINT DISPLACEMENT (CM   RADIANS)   STRUCTURE TYPE = SPACE

JOINT	LOAD	X-TRANS	Y-TRANS	Z-TRANS	X-ROTAN	Y-ROTAN	Z-ROTAN
2	7	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	8	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	10	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
	11	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
4	7	-0.1050	-0.0014	-0.0196	-0.0001	0.0083	0.0004
	8	-0.1233	-0.0014	-0.0196	-0.0001	0.0083	0.0005
	9	-0.1050	-0.0014	-0.0226	-0.0001	0.0083	0.0004
	10	-0.1123	-0.0014	-0.0196	-0.0001	0.0087	0.0005
	11	-0.1050	-0.0014	-0.0222	-0.0001	0.0083	0.0004
5	7	0.5162	0.0069	-0.0196	-0.0001	0.0083	0.0004
	8	0.4979	0.0069	-0.0196	-0.0001	0.0083	0.0005
	9	0.5162	0.0077	-0.0226	-0.0001	0.0083	0.0004
	10	0.5399	0.0069	-0.0196	-0.0001	0.0087	0.0005
	11	0.5162	0.0077	-0.0222	-0.0001	0.0083	0.0004
6	7	-0.7267	-0.0115	-0.0196	-0.0001	0.0083	0.0004
	8	-0.7450	-0.0115	-0.0196	-0.0001	0.0083	0.0005
	9	-0.7267	-0.0123	-0.0226	-0.0001	0.0083	0.0004

11	-0.7267	-0.0123	-0.0222	-0.0001	0.0083	0.0004

************** END OF LATEST ANALYSIS RESULT **************

82. PRINT SUPPORT REACTION ALL

SUPPORT REACTIONS -UNIT KN METE STRUCTURE TYPE = SPACE

JOINT	LOAD	FORCE-X	FORCE-Y	FORCE-Z	MOM-X	MOM-Y	MOM Z
2	7	5.88	33.44	0.00	7.84	-2.35	-20.55
	8	7.70	33.44	0.00	7.84	-2.35	-24.68
	9	5.88	33.44	1.21	9.95	-2.35	-20.55
	10	6.38	33.44	0.00	7.84	-2.47	-22.05
	11	5.88	33.44	0.50	9.34	-2.35	-20.55

************** END OF LATEST ANALYSIS RESULT **************

83. LOAD LIST 12 TO 17
84. PRINT MEMBER FORCES ALL

MEMBER END FORCES STRUCTURE TYPE = SPACE

ALL UNITS ARE -- KN METE (LOCAL)

MEMBER	LOAD	JT	AXIAL	SHEAR-Y	SHEAR-Z	TORSION	MOM-Y	MOM-Z
3	12	2	33.44	0.00	8.23	-3.29	-28.77	7.84
		4	-23.76	0.00	-8.23	3.29	0.00	-7.84
	13	2	46.82	0.00	8.23	-3.29	-28.77	10.98
		4	-33.26	0.00	-8.23	3.29	0.00	-10.98
	14	2	33.44	0.00	10.78	-3.29	-34.55	7.84
		4	-23.76	0.00	-9.00	3.29	0.00	-7.84
	15	2	33.44	1.69	8.23	-3.29	-28.77	10.80
		4	-23.76	0.00	-8.23	3.29	0.00	-7.84
	16	2	33.44	0.00	8.93	-3.46	-30.87	7.84
		4	-23.76	0.00	-8.73	3.46	0.00	-7.84
	17	2	33.44	0.70	8.23	-3.29	-28.77	9.94
		4	-23.76	-0.50	-8.23	3.29	0.00	-7.84
4	12	4	0.00	2.08	0.00	0.00	0.00	0.78
		5	0.00	0.00	0.00	0.00	0.00	0.00
	13	4	0.00	2.91	0.00	0.00	0.00	1.09
		5	0.00	0.00	0.00	0.00	0.00	0.00
	14	4	0.00	2.08	-0.38	0.00	0.14	0.78
		5	0.00	0.00	0.00	0.00	0.00	0.00
	15	4	0.00	2.08	0.00	0.00	0.00	0.78
		5	0.00	0.00	0.00	0.00	0.00	0.00
	16	4	0.00	2.08	-0.04	0.00	0.02	0.78
		5	0.00	0.00	0.00	0.00	0.00	0.00
	17	4	0.04	2.08	0.00	0.00	0.00	0.78
		5	0.00	0.00	0.00	0.00	0.00	0.00
5	12	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.00	21.68	-8.23	0.00	-3.29	-8.62
	13	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.00	30.35	-8.23	0.00	-3.29	-12.07

		4	0.00	21.68	-8.61	0.00	-3.44	-8.62
	15	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.00	21.68	-8.23	0.00	-3.29	-8.62
	16	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.00	21.68	-8.69	0.00	-3.47	-8.62
	17	6	0.00	0.00	0.00	0.00	0.00	0.00
		4	0.46	21.68	-8.23	0.00	-3.29	-8.62

************** END OF LATEST ANALYSIS RESULT **************

85. LOAD LIST ALL
86. PARAMETER 1
87. CODE BS5950
88. BEAM 2 ALL
89. CHECK CODE ALL

STAAD.Pro CODE CHECKING - (BSI)
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PROGRAM CODE REVISION V2.12_5950-1_2000

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/ FX	CRITICAL COND/ MY	RATIO/ MZ	LOADING/ LOCATION
3 ST	H428X407X20	PASS	ANNEX I.1	0.050	14
		33.44 C	34.55	7.84	-
4 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.002	13
		0.00	0.00	1.09	0.00
5 ST	H428X407X20	PASS	BS-4.2.3-(Y)	0.022	13
		0.00	-3.29	-12.07	0.75

************** END OF TABULATED RESULT OF DESIGN **************

90. FINISH

*********** END OF THE STAAD.Pro RUN ***********

**** DATE= JAN 20,2013   TIME= 10:10:56 ****

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*        For questions on STAAD.Pro, please contact        *
*   Research Engineers Offices at the following locations  *
*                                                          *
*              Telephone                 Email             *
*  USA:      +1 (714)974-2500       support@bentley.com    *
*  CANADA    +1 (905)632-4771       detech@odandetech.com  *
*  UK        +44(1454)207-000       support@bentley.com    *
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